MIL-F-52345C (ME)
4 August 1989
SUPERSEDING
MIL-F-52345B (ME)
21 November 1980

MILITARY SPECIFICATION

FLOW CONTROL KIT, HOSELINE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the components of a kit to control the flow of fuel in a hoseline system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

GGG-T-1310	–	Tape, Fish, Steel.
PPP-B-601	–	Boxes, Wood, Cleated-Plywood.
PPP-T-60	–	Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3835

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MIL-P-116	–	Preservation, Methods of.
MIL-B-121	–	Barrier Material, Greaseproofed, Waterproof, Flexible.
MIL-T-704	–	Treatment and Painting of Materiel.
MIL-A-8625	–	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-10387	–	Couplings, Clamp, Pipe: With Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe.
MIL-P-10388	–	Pipe Fittings: One or More Ends Grooved.
MIL-C-27487	–	Coupling Halves, Quick-Disconnect, Cam-Locking Type.
MIL-H-52262	–	Hose, Assembly, Rubber: Lightweight, Collapsible, 4-Inch.
MIL-V-58039	–	Valves, Gate, Rising Stem, Double Acting Aluminum.

STANDARDS

MILITARY

MIL-STD-105	–	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	–	Marking for Shipment and Storage.
MIL-STD-130	–	Identification Marking of US Military Property.
MIL-STD-889	–	Dissimilar metals.
MIL-STD-1472	–	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1190	–	Minimum Guidelines for Level C Preservation, Packing and Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Form Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

13204E9839

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the US Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSH, Fort Belvoir, VA 22060-5606.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Description. The flow control kit (hereinafter called "kit"), shall consist of the following components contained in a steel chest as specified herein:

Components	Quantity
Gate valves	2
Tee	1
Couplings, clamp, pipe	8
Check valves	2
Hose assemblies	2
Strainer assembly	1
Couplings, grooved-to-external-thread	2
Coupling, Grooved-to-internal-thread	2
Coupling halves, male, external pipe thread	2
Coupling halves, female, internal pipe thread	2
Tape, electrician's, fishing	1

3.1.1 Drawing. The drawing forming a part of this specification is an engineering design drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contract officer. Any data (e.g. shop drawings, layouts, flow sheets, and processing procedures) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subject to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawing. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 3.3.2).

3.3.1 Material deterioration prevention and control. The kit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the kit may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieced and parts

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incorporated in the kit may be newly fabricated from recovered materials to the maximum extent practicable, provided the kit produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the kit.

3.4 Gate valves. Gate valves shall be aluminum and shall conform to MIL-V-58039, type I, 4-inch (101.6 mm), except that the anodizing and color shall be as specified in 3.14. Each end of the valves shall be equipped with a 4-inch (101.6 mm) aluminum coupling conforming to MIL-P-10388. Rotary valve controls shall be provided with double-ended arrows showing direction of operation and labeled at each end to indicate the functional result (open, close).

3.5 Tee. The tee shall conform to MIL-P-10388, 4-inch (101.6 mm), all ends shall be grooved, and shall be aluminum.

3.6 Couplings (clamp, pipe). Couplings shall conform to MIL-C-10387, 4-inch (101.6 mm), and shall be aluminum.

3.7 Check valves. Check valves shall be 4-inch iron and conform to Wheatley Pump and Valve, Inc., P/N 830-041030-001 or equal; working pressure shall be 300 pounds per square inch (psi) (2068 kPa). The ends shall be grooved as specified in MIL-P-10388. The valves (except the grooves) shall be treated and painted conforming to MIL-T-704, type F or G, color green 383, number 34094.

3.8 Hose. The hose length shall be 5 feet (1524 mm) +2 inches (50.8 mm), less fittings and shall be provided with an adapter at each end. The hose and adapters shall conform to MIL-H-52262. Hose furnished to the Government shall be not more than 12 months old from date of cure of the rubber to date of acceptance by the Government.

3.9 Strainer assembly. The strainer assembly shall conform to Drawing 13204E9839, and as specified herein.

3.10 Couplings (grooved to threaded). Couplings shall conform to MIL-P-10388, aluminum, 4-inch (101.6 mm), internal threaded on one end, and external threaded on one end. External threaded couplings shall be not more than 6 inches (152.4 mm) long.

3.11 Coupling halves. Coupling halves shall conform to MIL-C-27487, class 1, 4-inch (101.6 mm). Male, external pipe thread shall conform to type III, and female, internal pipe thread shall conform to type IV.

3.12 Tape. The tape shall conform to GGG-T-1310, type I, style 1, size 6; the reel shall be metal.

3.13 Chest. A steel chest shall be provided for the components of the kit. The chest shall be not more than 22 inch (558.8 mm) in height, and shall be constructed of steel not less than 0.073 inch (1.85 mm) thick. The chest shall have a hinged cover and latches to secure the cover in the closed position. The cover shall be self-supporting in the open position. Handles shall conform to MIL-STD-1472, 5.9.11.5. The sides and top of the chest shall be labeled "MECHANICAL LIFT

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ONLY" in one inch (25.4 mm) high letters. The chest shall be provided with mechanical lift provisions (i.e., forklift pockets, lift eye, etc.) The mechanical lift provisions shall be labeled (i.e., "FORKLIFT HERE", "LIFT EYE", etc.) as appropriate. The chest cover and front shall be labeled "HOSELINE FLOW CONTROL KIT in one inch (25.4 mm) high letters. Bins, cradles, or blocking shall be provided for the components of the kit to prevent damage during shipment and handling. Break all exposed sharp edges and corners and deburr.

3.13.1 Loading plan. A loading plan indicating the location of all items in the chest shall be permanently attached to the inside of the cover. The loading plan shall provide pictorial verbal instruction for loading the chest. Lettering shall comply with MIL-STD-1472, 5.5.5.

3.13.2 Treatment and painting. The chest shall be cleaned, treated and painted conforming to MIL-T-704, type F or G, color green 383, number 34094.

3.14 Anodizing. All aluminum components shall be anodized conforming to MIL-A-8625, type II, to a minimum thickness of 0.0004 inch (0.01 mm) for aluminum castings and 0.0007 inch (0.02 mm) for aluminum wrought alloys, and painted green 383, color number 34094.

3.15 Identification marking. The kit identification shall conform to MIL-STD-130. Identification marking shall be applied to the outside of the cover of the chest.

3.16 Workmanship. All components of the kit and chest shall be free of harmful extraneous material such as sand, dirt, sprues, scale and flux. External surfaces shall be free from burrs, sharp edges, and corners.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, and examined to conform to the referenced specifications and standards as applicable.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1, in the order shown. The presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall conform to MIL-STD-105. The sample unit for this inspection shall be one complete flow kit.

4.4.2 Examination. Samples selected shall conform to 4.4.1 and shall be examined for major and minor defects specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The kit shall be examined as specified herein for the following defects:

Number	Defects	Requirement Paragraph
Major		
101.	Components of the kit not as specified.	3.1
102.	The number of components not as specified.	3.1
103.	Drawing not furnished.	3.1.1
104.	Material not as specified.	3.3
105.	Material not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
106.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
107.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.1.2
108.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the kit.	3.3.2
109.	Gate valves not as specified.	3.4

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110.	Tee valves not as specified.	3.5
111.	Couplings (clamp, pipe) not as specified.	3.6
112.	Check valves not as specified.	3.7
113.	Hose not as specified.	3.8
114.	Strainer assembly not as specified.	3.9
115.	Couplings (grooved to threaded) not as specified.	3.10
116.	Coupling halves not as specified.	3.11
117.	Tape not as specified.	3.12
118.	Chest not as specified.	3.13
119.	Loading plan missing or not as specified.	3.13.1
120.	Workmanship not as specified.	3.16

Minor

201.	Treatment and painting not as specified.	3.13.2
202.	Color not as specified.	3.13.2, 3.14
203.	Anodizing not as specified.	3.14
204.	Identification marking missing, incomplete or illegible.	3.15

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall conform to MIL-STD-105.

4.6.1.3 Examination. Samples selected shall conform to 4.6.1.2 and shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

121. Unpainted metal surfaces of components requiring a contact preservative not coated as specified.
122. Land-sealing surface of grooved-end fittings not completely covered with barrier material and tape, as specified in 5.1 and as shown on figure 1.
123. Chest lid not sealed with tape as specified.
124. Shipping containers not as specified for levels A or B.
125. Marking missing, illegible, incorrect or incomplete for level A, level B, or C.
126. Packing not as specified for level A, B, or C.

5. PACKAGING

5.1 Preservation (see 6.5). Unpainted metal surfaces of components of the kit requiring the application of a contact preservative conforming to MIL-P-116, shall be coated with type P-1 or, as applicable, P-19 preservative. The preservatives shall conform to the applicable specifications listed in MIL-P-116 and shall be applied conforming to MIL-P-116. The land-sealing surface of

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grooved end fittings shall be completely covered with a single wrap of barrier material conforming to MIL-B-121, type II, class 2 (see figure 1). The wrap for the land-sealing surface shall be flush with the end of the fitting and shall extend back beyond the groove not less than 0.250 inch (6.35 mm). The wrap of the barrier material shall be covered with a single wrap of tape conforming to PPP-T-60, type IV. The tape shall be of a width that will completely cover the barrier material and shall extend not less than 0.125 inch (3.18 mm) beyond the end of the fitting and extend back beyond the barrier material not less than 0.500 inch (12.7 mm). The end of the tape extending beyond the end of the fitting, and the tape extending beyond the wrap shall be rolled in place against the end wall of the fitting and against the fitting (see figure 1). The preserved components comprising a complete kit shall then be placed and secured in the chest (see 3.13) conforming to the loading plan (see 3.13.1). The chest cover shall then be closed and latched in the closed position with the provided fastening/latching devices (see 3.13). After the cover has been latched, the seam between the chest cover and the chest body shall be completely sealed with tape conforming to PPP-T-60, type IV.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each complete kit, preserved and secured within the chest as specified in 5.1, shall be individually packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be as specified in the box specification, except that strapping shall be flat and the finish B.

5.2.2 Level B. Each complete kit, preserved and secured within the chest as specified in 5.1, shall be packed as specified in 5.2.1 for level A, except that boxes shall be domestic type and the strapping may be finish A.

5.2.3 Level C. Each complete kit, preserved and secured within the chest as specified in 5.1, shall be individually packed conforming to MIL-STD-1190.

5.3 Marking. In addition to any special or identification markings required by the contract or purchase/delivery order (see 6.2), each container, excluding the chest, shall be marked conforming to MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The kit is intended for use in controlling the flow in a 4.00-inch (101.6 mm) lightweight, collapsible, hoseline fuel system.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1).

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c. Timeframe required for submission of first article and number of kits required (see 3.2).
d. When the Government will conduct any or all of the first article examination. When the Government will conduct some but not all of the first article examination, the contracting officer should specify which examination will be conducted by the Government and which examination will be conducted by the contractor (see 3.2).
e. Level of packing required (see 5.2).
f. Any special marking (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a first production item. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Flow control kit, hoseline
Hoseline, flow control kit
Kit, hoseline, flow control

6.5 Levels of protection. For the purpose of marking, the levels of preservation and packing shall be designated A/A, A/B, or A/C.

6.6 AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/Lot Tolerances Percent Defectives) from military specifications.

(Certified by/date: _____)

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

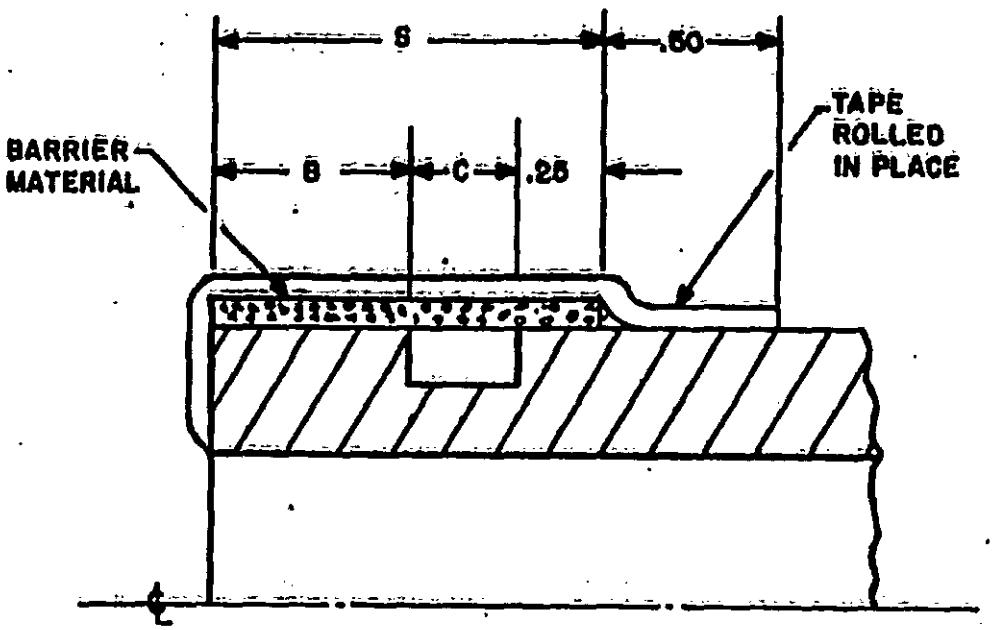
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DLA – CS

Project 3835-A087

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S ——— LAND SEALING SURFACE

B ——— LAND

C ——— GROOVE

NOTE :

1. ALL DIMENSIONS ARE IN INCHES.

FIGURE 1. Land sealing surface for grooved end pipe, tube end fittings.

X-449C

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-52345C	890804

3. DOCUMENT TITLE

Flow Control Kit, Hoseline

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

Reinstatement

5. REASON FOR RECOMMENDATION

Reinstatement in order to convert to a performance specification.

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
Mitchell, Elizabeth C.	US ARMY TACOM-TARDEC	
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command AMSTA-TR-D/210 Warren, MI 48397-5000	d. TELEPHONE (Include Area Code) (1) Commercial (810) 574-4154 (2) AUTOVON (If applicable) 786-4154	7. DATE SUBMITTED (YYMMDD) 980910

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Mitchell, Elizabeth C.	(1) Commercial (810) 574-4154	(2) AUTOVON 786-4154
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command AMSTA-TR-D/210 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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